

csiro-biomass

10th Place Solution

Rank	10
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Version	Original

Source: <https://www.kaggle.com/competitions/csiro-biomass/discussion/671053>

Retrieved with Kaggle CLI on 2026-07-18.

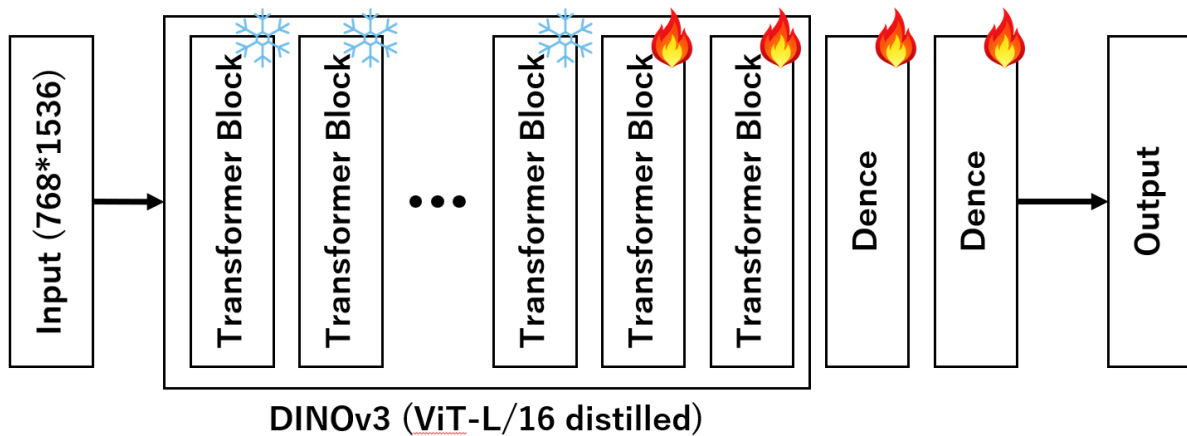
We would first like to express our gratitude to the competition host and the Kaggle staff for organizing this outstanding competition. Below, we introduce the my solution

Solution

Item	Description
Model	DINOv3 ViT-L/16 distilled (Fine-tuned: regression head + last two transformer blocks)
Input	Images resized to 768×1536
Output	5 target variables + auxiliary-loss features (Height, NDVI, and whether Species is Ryegrass (Ryegrass_Clover))
Loss function	Huber loss weighted the same way as the evaluation metric + auxiliary loss
Preprocessing	- Removed 8 samples suspected as label noise after inspection - Standardized the target variables
CV	Grouped by State - Sampling-Date, then manually reviewed and merged similar groups. Split into 5 folds using a greedy method so that each target variable's distribution is similar across folds. CV and LB seemed correlated, so I tried to improve both.
Anti-shake measures	Data augmentation to reproduce unnatural artifacts present in the train set (red-dot artifacts, and a date stamp in the bottom-right corner)
Ensemble	Trained 3 models (with/without auxiliary losses) and averaged their predictions
Other	- EMA - TTA (horizontal flip)

1. Model, Input/Output, and Ensemble

I used DINOv3 ViT-L/16 distilled. Images were resized to 768×1536 and fed into the model. The model outputs consist of the five target variables plus optional auxiliary-loss features, and the final prediction is the average of three models with different output configurations.



Pattern	Output configuration
1	5 target variables
2	5 target variables + Height + NDVI
3	5 target variables + Height + NDVI + Species (binary classification: whether it is Ryegrass (Ryegrass_Clover))

2. Preprocessing

For samples with State=WA and Sampling_Date=2015-08-21, the label indicates Dry_Green_g = 0. However, upon inspecting the images, there were clearly plants other than Clover present. I therefore treated these as label errors and removed the following 8 samples from the training data:

- ID681680726.jpg
- ID697718693.jpg
- ID40849327.jpg
- ID230058600.jpg
- ID1403107574.jpg
- ID1337107565.jpg
- ID1139918758.jpg
- ID1761544403.jpg

This improved the score by +0.01 on CV, +0.01 on Public, and +0.01 on Private. (I corrected it because I had made a mistake.)

3. Anti-shake Measures

The training data contains unnatural artifacts such as the following, so I applied data augmentation to reproduce them:

- Red-dot artifacts, likely caused by flash
- The capture date/time appearing in the bottom-right corner of the image

This did not help on CV or Public, but improved the Private score by +0.01. When I tested them separately, the date/time augmentation did not improve the Public score, so this suggests the private set may contain some red-dot artifacts.

4. Tried but Not Effective

- Manifold Mixup
- Tuning all layers with LoRA
- Pseudo-labeling using data that was not used for training
- Using patch tokens (maybe I just used it incorrectly?)
- Using PlantCLEF2024 data / a pre-trained model (DINOv2)

In Closing

If you have any questions, feel free to ask. Thank you for reading!

Supplemental Q&A

Geremie Yeo · 2026-01-31T05:59:52.243000

For samples with State=WA and Sampling_Date=2015-08-21, the label indicates Dry_Green_g = 0. However, upon inspecting the images, there were clearly plants other than Clover present. I therefore treated these as label errors and removed the following 8 samples from the training data:

what was your OOF correlation between the model trained with/without these 8 samples? Did the same and for me was 0.99x, extremely similar...

owaowata · 2026-01-31T15:03:30.867000

The correlation between the predicted Dry_Total_g values from the model trained with the 8 samples and the model trained without them was 0.897. Also, I wrote in my solution that the CV score improved by 0.04, but that was a calculation mistake—the correct improvement was 0.01.